FROB C5 exact quotient matrix factor kernels

Validation audit, 22 October 2026

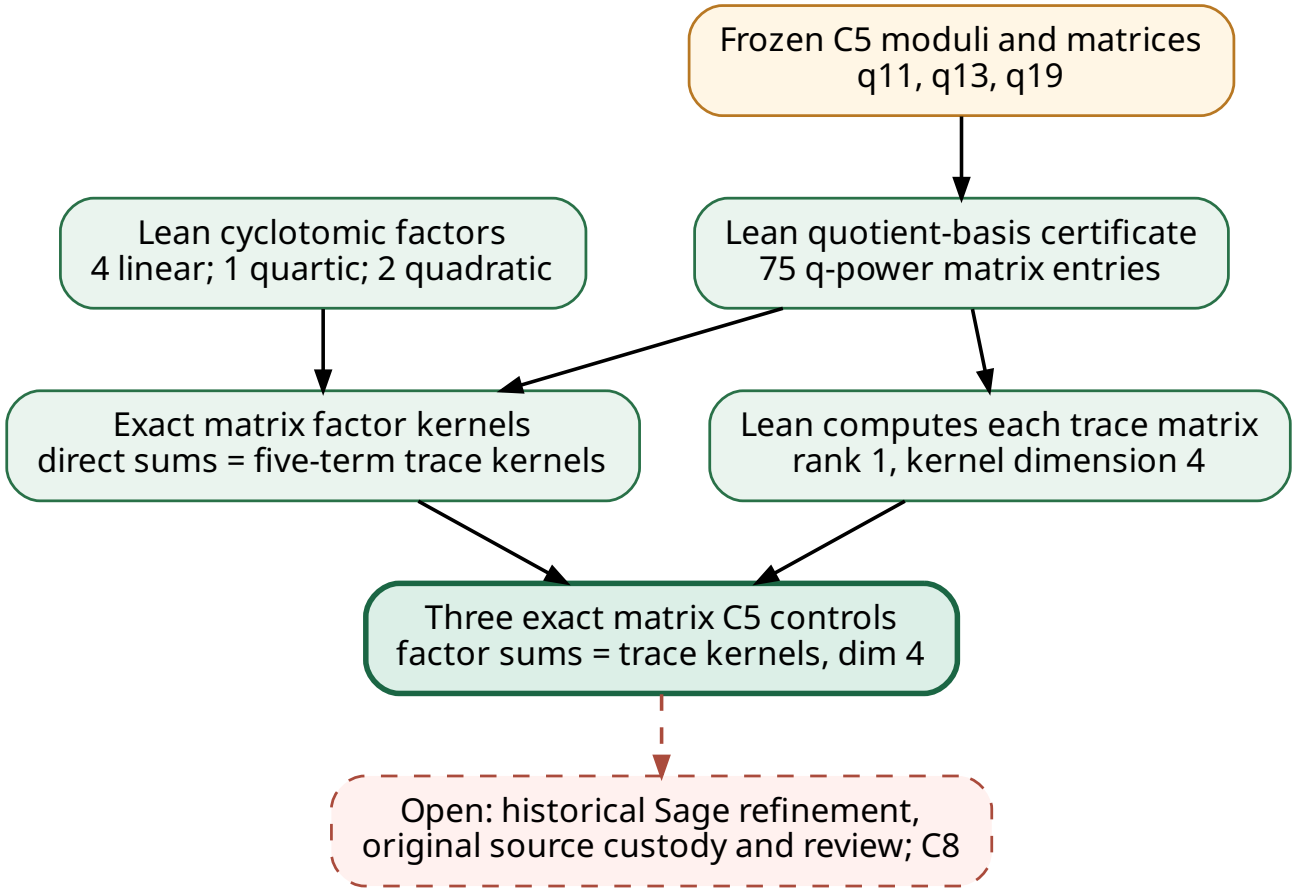
Scope. The frozen `RUN-FROB-91ee9c-7119f2` core cells have three previously certified 5×5 quotient-basis q-power matrices. This target evaluates their nontrivial cyclotomic factor polynomials and their five-term trace polynomials as linear maps over the base fields.

Base field	Nontrivial factors	Lean matrix result	Trace-kernel dimension
$\mathbf{F}_{11}$	four distinct linears	successive direct sum $= \ker T$	4
$\mathbf{F}_{13}$	one irreducible quartic	factor kernel $= \ker T$	4
$\mathbf{F}_{19}$	two irreducible quadratics	direct sum $= \ker T$	4

Proof. The exact matrix entries were proved from frozen quotient moduli in a separate Lean certificate. The present source computes $M_q^5 = I$ and $T_q = I + M_q + M_q^2 + M_q^3 + M_q^4$ for each q. It factors each trace matrix as a rank-one diagonal matrix times an invertible upper-triangular matrix. Rank-nullity gives $\dim \ker T_q = 4$. Previously checked polynomial factorizations and coprimality give the direct factor-kernel sums, instantiated here with `Matrix.toLin'` of the exact frozen matrices. All three statements are combined in one target theorem.

Machine verification. Lean source commit `458c91dbdcb8` uses pinned Lean 4.35.0-rc1. The exact-source verifier passed an 8,973-job full build and 3,097-job target build. Its global audit found 647 theorem declarations with standard axioms only; the requested theorem audit passed. Independent semantic review remains pending.

Boundary. This closes the mathematical C5 factor-kernel calculation for the three exact recorded matrices. A full-map q-power representation lemma, historical Sage implementation refinement, original run source custody, independent code review, C8 and the full experiment success gate remain open. No experiment was rerun and no performance dashboard value changed.


Frozen C5 moduli and matrices
q11, q13, q19
Lean cyclotomic factors
4 linear; 1 quartic; 2 quadratic
Lean quotient-basis certificate
75 q-power matrix entries
Exact matrix factor kernels
direct sums = five-term trace kernels
Lean computes each trace matrix
rank 1, kernel dimension 4
Three exact matrix C5 controls
factor sums = trace kernels, dim 4
Open: historical Sage refinement,
original source custody and review; C8